# Exercise Sheet 22

The notation is that of Lecture 22: $V = \mathbb{C}^{2n}$ with skew form $\beta$, basis $(e_1, \dots, e_n, f_1, \dots, f_n)$, $\mathfrak{g} = \mathfrak{sp}_{2n}$ and weights written as $(a_1, \dots, a_n) = \sum a_i \varepsilon_i$. These are $\varepsilon$-coordinates, not Dynkin labels. The invariant tensor is $\Omega = \sum_i (e_i \otimes f_i - f_i \otimes e_i)$. Exercises 22.1–22.4 are routine. Exercises 22.5–22.7 transfer the methods.

**Exercise 22.1 (The adjoint module as a symmetric square).** For $u, v \in V$ define $X_{uv} \in \mathfrak{gl}(V)$ by

$$X_{uv}(x) = \beta(u, x)\, v + \beta(v, x)\, u.$$

(i) Prove that $X_{uv} \in \mathfrak{g}$.

(ii) Prove that $[A, X_{uv}] = X_{Au,v} + X_{u,Av}$ for $A \in \mathfrak{g}$.

(iii) Deduce that $uv \mapsto X_{uv}$ is an isomorphism $\operatorname{Sym}^2 V \to \mathfrak{g}$ of $\mathfrak{g}$-modules.

**Exercise 22.2 (Orthogonal versus symplectic contractions).** Let $N \geq 3$ and let $W = \mathbb{C}^N$ carry a nondegenerate symmetric form $\gamma$. Put

$$\mathfrak{k} = \mathfrak{so}_N = \{X \in \mathfrak{gl}(W) : \gamma(Xu, v) + \gamma(u, Xv) = 0 \text{ for all } u, v \in W\}.$$

Let $\kappa_\gamma : W \otimes W \to \mathbb{C}$ and $\kappa_\beta : V \otimes V \to \mathbb{C}$ be the linear maps with

$$\kappa_\gamma(u \otimes v) = \gamma(u, v) \quad \text{and} \quad \kappa_\beta(u \otimes v) = \beta(u, v).$$

(i) Show that $\kappa_\gamma$ vanishes on $\Lambda^2 W$ but not on $\operatorname{Sym}^2 W$. Show the reverse for $\kappa_\beta$.

(ii) Let $\operatorname{Sym}^2_0 W = \ker \kappa_\gamma \cap \operatorname{Sym}^2 W$. Prove $W \otimes W = \Lambda^2 W \oplus \operatorname{Sym}^2_0 W \oplus L$ as $\mathfrak{k}$-modules, where $L$ is a trivial line. Give the dimensions.

(iii) Compare $N = 5$ with $\mathfrak{sp}_4$. Where does the invariant line lie in each case?

(iv) Show that symmetric tensors need not be harmonic for $\gamma$.

**Exercise 22.3 (Highest vectors in $\mathfrak{sp}_4$ products).** Let $n = 2$. Let $X_1 \in \mathfrak{gl}(V)$ satisfy

$$X_1 e_2 = e_1, \qquad X_1 f_1 = -f_2, \qquad X_1 e_1 = X_1 f_2 = 0.$$

(i) Show that $X_1 \in \mathfrak{g}$ and that $X_1$ has weight $\alpha_1 = \varepsilon_1 - \varepsilon_2$.

(ii) Find a highest vector of weight $(2, 1)$ in $V \otimes \operatorname{Sym}^2 V$.

(iii) Find one in $V \otimes \Lambda^2_0 V$.

**Exercise 22.4 (The multiplicity space as an $S_3$-module).** Let $n = 2$, and let $\iota_{12}, \iota_{13}, \iota_{23}$ be the basis of $M = \operatorname{Hom}_{\mathfrak{g}}(V, V^{\otimes 3})$ from Lecture 22. Let $S_3$ act on $M$ by $\varphi \mapsto \sigma \circ \varphi$.

(i) Compute the matrices of $(1\,2)$ and $(1\,2\,3)$.

(ii) Decompose $M$ into irreducible $S_3$-modules.

**Exercise 22.5 (A product for $\mathfrak{sp}_6$).** Let $n = 3$. Prove

$$V \otimes L(\omega_2) \cong L(\omega_1 + \omega_2) \oplus L(\omega_3) \oplus L(\omega_1), \qquad 84 = 64 + 14 + 6.$$

**Exercise 22.6 (The length hypothesis for $\Lambda^3\mathbb{C}^4$).** Let $n = 2$ and define

$$c_3(x \wedge y \wedge z) = \beta(x,y)z - \beta(x,z)y + \beta(y,z)x.$$

(i) Show that $c_3$ is a well-defined $\mathfrak{g}$-map $\Lambda^3 V \to V$.
(ii) Prove that $c_3$ is an isomorphism, and deduce $\Lambda^3_0 V = 0$.
(iii) Show that $v \mapsto c_3(\Omega_\wedge \wedge v)$ is the identity, where $\Omega_\wedge = e_1 \wedge f_1 + e_2 \wedge f_2$.

**Exercise 22.7 (Which tensor powers contain $L(\lambda)$).** Let $\lambda \in P_+$ for $\mathfrak{sp}_{2n}$.

(i) Prove that $L(\lambda)$ occurs in $V^{\otimes d}$ if and only if $d \geq |\lambda|$ and $d \equiv |\lambda| \pmod 2$.
(ii) For $\mathfrak{sp}_4$, list the $d \leq 5$ for which $L(\omega_1 + \omega_2)$ occurs in $V^{\otimes d}$.